Authors' Responses

Journal Title:	IEEE Robotics and Automation Letters
Reference Number:	26-3841v1
Title of Paper:	VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
Authors:	Anonymous Authors

We would like to extend our heartfelt gratitude to the Editor-in-Chief and the Associate Editor for their time and effort in handling our paper. We also sincerely thank the two anonymous reviewers for their invaluable comments and suggestions, which have greatly helped us improve the quality of our work.

We have completed a major revision of the manuscript and expanded its technical explanations in the following four main areas:

1) validated the necessity and contribution of VLM guidance by adding two ablations that isolate its visual and semantic components and by extending comparisons with fixed and adaptive intervention strategies;
2) improved reproducibility by providing the exact VLM prompt, model and inference settings, visual-input selection procedure, output processing, update frequency, resource usage, failure handling, and relevant training details;
3) strengthened the evaluation of the proposed Bellman success-value critic by reporting task-outcome prediction accuracy, unsafe-event detection, and false-negative rates under student-controlled rollouts; and
4) expanded the real-world evaluation by increasing the number of trials, reporting results for all 22 objects and pooled seen/unseen confidence intervals, adding a physical baseline comparison, and clarifying the empirical scope and limitations of our safety and deployment claims.

The detailed responses to the reviewers' comments are attached. We would be happy to incorporate any further feedback or suggestions into the final version of the manuscript.

RESPONSE TO REVIEWER #14

We sincerely thank the reviewer for the careful assessment of our work and for the constructive suggestions. Below are our point-by-point responses.

Comment 1: ***whether the VLM and visual information provide a clear advantage over simpler adaptive approaches.***

[Response:] We have added ablations (g) and (h) to isolate the contribution of the VLM to the adaptive intervention pipeline. We have also expanded the discussion of Table III in the Results section.

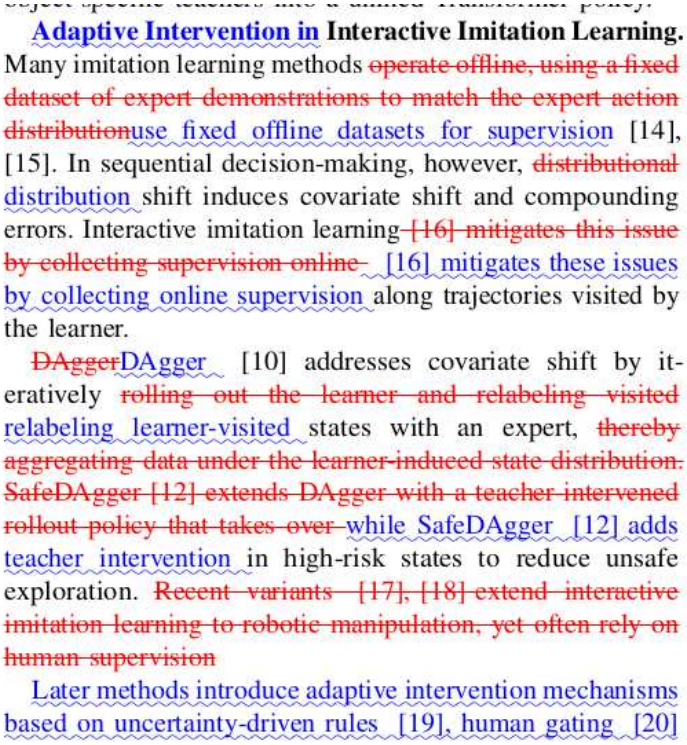
Table 1: Performance of ablations (a)–(h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5	72.7	26.5
(b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0	72.3	25.8
(e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(g)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(h)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

The results support our claim that VLM-based intervention outperforms both LLM-based intervention and performance-based curriculum adaptation. Semantic information alone reduces the unsafe exploration rate, while combining visual and semantic information enables more precise intervention, achieving the highest task success rate and the lowest unsafe exploration rate. Although the performance-based curriculum reduces unsafe exploration to some extent, it sacrifices task success, suggesting that a simple rule-based method is insufficient for guiding intervention policies in complex scenarios. The detailed design of the performance-curriculum ablation (h) is provided in Supplementary Section C.5.

Comment 2: ***The related work on adaptive intervention should also be expanded.***

[Response:] We have reformulated the Related Work section to more clearly discuss adaptive intervention methods. In particular, AIM provides an important foundation for our work, and we have added it to the reference list. Compared with prior methods such as HG-DAgger and ThriftyDAgger, our proposed pipeline does not require human intervention, enabling fully autonomous intervention during online policy distillation.



Adaptive Intervention in Interactive Imitation Learning. Many imitation learning methods ~~operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution~~use fixed offline datasets for supervision [14], [15]. In sequential decision-making, however, ~~distributional~~ distribution shift induces covariate shift and compounding errors. Interactive imitation learning ~~[16] mitigates this issue by collecting supervision online.~~ [16] mitigates these issues by collecting online supervision along trajectories visited by the learner.

~~DAgger~~DAgger [10] addresses covariate shift by iteratively ~~rolling out the learner and relabeling visited~~ relabeling learner-visited states with an expert, ~~thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over~~ while SafeDAgger [12] adds teacher intervention in high-risk states to reduce unsafe exploration. ~~Recent variants [17], [18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision~~

Later methods introduce adaptive intervention mechanisms based on uncertainty-driven rules [19], human gating [20]

Comment 3: ***the authors should provide the VLM prompt and explain how the images are selected and how the VLM outputs are used.***

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt and representative visual input examples are included in Supplementary Section B.

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "delta_l2_threshold": <number>, "delta_success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
User message:
You are a conservative threshold advisor for
DexSafeDagger robot distillation.
Do not decide intervention directly.  Recommend
threshold displacements for the existing gates.
Task:  Propose displacements for the
teacher-student disagreement and predicted-success
thresholds used in dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys:
delta_l2_threshold, delta_success_threshold,
confidence, reason.
Keep changes conservative.  Prefer small threshold
shifts unless the recent window shows clear over-
or under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min},
{success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
Recent rollout statistics:
{
 "frame":  {frame}, "intervention_rate":  {beta},
"step":  {step},
 "sample_count":  {N}, "l2_threshold":
{tau_delta},
 "success_threshold":  {tau_Q},
"unsafe_episode_rate":  {r_unsafe},
 "l2":  {"min":  {v}, "max":  {v}, "mean":  {v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "success":  {"min":  {v}, "max":  {v}, "mean":
{v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "unsafe_reason_prop":  { "hand_too_far":  {v},
     "harmful_collision":  {v},
"object_out_of_bound":  {v},
     "palm_flipped":  {v}},
 "visual_buffer_size":  {B}, "visual_samples_attached":
{M},
 "visual_samples":  [{ "env_id":  {v}, "frame":
{v},
     "image_attached":  true, "image_key":
{v}, "source":  {high_l2 | low_success |
unsafe_triggered},
     "l2_threshold":  {v}, "success_threshold":
{v},
     "teacher_student_l2":  {v},
"predictor_success":  {v},
     "object_id":  {v}, "object_name":  {v},
"object_env_count":  {v},
     "step":  {v}, "unsafe":  {true | false}}, ...]

}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata:  {the corresponding
record above, excluding "image_attached"}
<IMAGE {i}:  JPEG data URL, detail="high">
```

Comment 4: ***The Bellman success-value critic is also important for intervention, but its performance is not evaluated in detail. Some evaluation of the critic, especially its errors for unsafe cases, would be helpful.***

[Response:] We use the resulting student and fixed-threshold Bellman critic from ablation (e) to evaluate the critic's performance. Both networks are frozen during evaluation, and the critic runs in parallel as a passive evaluator. We evaluate 200 episodes for each of the 40 training objects in simulation. Table 2 reports the observed lift-success and unsafe-episode rates together with the task-outcome prediction accuracy, the

unsafe-episode detection rate, and its corresponding false-negative rate.

The lift-success rate and unsafe-episode rate characterize the student policy's ability to accomplish the task and the frequency of unsafe outcomes, respectively. Task-outcome prediction accuracy is the percentage of episodes for which the critic correctly predicts the task outcome. The unsafe-episode detection rate is the percentage of unsafely terminated episodes that are flagged by the critic before termination. The unsafe false-negative rate is the complementary percentage of unsafe episodes that the critic does not flag before termination.

Table 2: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

Comment 5: ***Please also clarify the number of runs/random seeds used for the results.***

[Response:] We run 30 real-world evaluations for each object, distributed equally across random seeds 40, 41, 42, 43, and 44 (six evaluations per seed).

Comment 6: ***Finally, I am not fully convinced by the use of the term "safety-critical." The current safety evaluation is based on task-specific unsafe conditions rather than formal safety guarantees. The authors should clarify the scope of this claim.***

[Response:] We acknowledge that the term "safety-critical" is too strong for an adaptive intervention pipeline such as ours. We have therefore revised the title to "VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation" and removed or qualified the term elsewhere in the manuscript. The revised text explicitly states that our method provides empirical risk reduction under task-specific event definitions rather than a formal safety guarantee, and that safety violations remain possible.

Comment 7: ***However, only VLM-DexSafeDagger evaluated on the real robot does not sufficiently prove the proposed concept.***

[Response:] We deploy the frozen RGB student policies resulting from ablations (e) and (f) on the real robot using the same object set, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates for the seen and unseen object sets in Main Paper Table IV.

Table 3: Overall real-world comparison of the frozen student policies resulting from ablations (e) and (f). Rates are reported separately for the 14 seen and 8 unseen objects.

Metric (%)	Ablation (e)		Ablation (f)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

Only the resulting student policy is deployed; the teacher, Bellman critic, VLM, and threshold-adaptation mechanism are not active during physical evaluation.

Comment 8: ***It would be helpful to provide overall results for all 22 objects, for example by reporting results separately for seen and unseen objects.***

[Response:] We have doubled the number of evaluation episodes for each object and reported the per-object and aggregate seen/unseen results for all 22 objects in Supplementary Tables IV and V.

Table 4: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 5: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

As recommended, we have also extended Tables II–IV in the main paper to report aggregate results for each

evaluation setting.

RESPONSE TO REVIEWER #22

We sincerely thank the reviewer for the detailed and constructive comments. Below are our point-by-point responses.

Comment 1: ***The term "safety-critical" is too strong because the framework supplies empirical intervention rather than formal constraint satisfaction. State explicitly that safety violations remain possible.***

[Response:] We acknowledge that the term "safety-critical" is too strong for an adaptive intervention pipeline such as ours. We have therefore revised the title to "VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation" and removed or qualified the term elsewhere in the manuscript. The revised text explicitly states that our method provides empirical risk reduction under task-specific event definitions rather than a formal safety guarantee, and that safety violations remain possible.

Comment 2: ***Demonstrate that the VLM is necessary by comparing against fixed thresholds, rule-based adaptation, Bayesian optimization, curriculum schedules, grid search, and a small learned threshold network.***

[Response:] We have supplemented the ablation study with variants (g) and (h). Together with variants (e) and (f), these experiments form a controlled ladder of adaptive intervention strategies: fixed thresholds, a performance-based curriculum, semantics-guided intervention-policy adaptation, and visual-semantic-guided adaptive intervention. We have added a corresponding discussion to the Results section. Table III shows that adaptive intervention reduces the unsafe rate relative to fixed thresholds. However, the curriculum-based method sacrifices lift success, whereas incorporating semantic information recovers task performance while further reducing unsafe exploration. Overall, combining visual and semantic information outperforms the other approaches, achieving the best balance between lift success and unsafe rate.

The detailed design of the performance-curriculum ablation (h) is provided in Supplementary Section C.5.

Table 6: Performance of ablations (a)–(h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5	72.7	26.5
(b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0	72.3	25.8
(e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(g)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(h)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

Comment 3: ***The Bellman critic predicts combined "safe-lift-success," thereby conflating task success***

with safety. Separate safety risk and task value, and report calibration, false-negative rates, and reliability diagrams.

[Response:] Lift success and unsafe termination measure different aspects of an episode and are not mutually exclusive. Lift success is recorded once the object satisfies the prescribed height-and-hold criterion, but satisfying this criterion does not terminate the rollout. The episode continues to be monitored for unsafe events until timeout or unsafe termination. Consequently, an episode may achieve lift success and subsequently terminate because of an unsafe event. The Bellman critic is therefore trained on the stricter joint safe-lift-success target, which is positive only when the lift criterion is satisfied without an unsafe event.

We evaluate the resulting student and fixed-threshold Bellman critic from ablation (e). Both networks are frozen, teacher intervention is disabled, and the critic runs in parallel as a passive evaluator. We separately report task success and unsafe termination outcomes, including the critic's unsafe-episode detection and false-negative rates. The results are reported in Supplementary Section C.1.

Table 7: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

Comment 4: ***Teacher intervention is assumed safe, yet teacher policies may collide or fail. Quantify teacher unsafe rates and define how the framework responds when teacher and critic disagree.***

[Response:] Teacher takeover is not guaranteed to be safe; it provides an empirical tendency toward safer exploration rather than a formal guarantee. We report the teacher policies' unsafe rates in Table III of the main paper and, in this revision, provide mean statistics across all tested objects. The intervention gate follows a conservative OR rule: intervention is triggered if either the teacher–student action disagreement exceeds its threshold or the critic's predicted success falls below its threshold. Thus, when the two signals disagree, the conservative outcome is to intervene. Comparing ablations (c) and (e) shows that adding the critic-based branch substantially reduces the unsafe exploration rate.

Comment 5: ***Provide exact VLM prompts, model version, decoding settings, image-selection procedure, output parsing, update frequency, latency, cost, and failure handling. Reproducibility may otherwise depend on an external proprietary service.***

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt, representative visual-input examples, model and inference configuration, image-selection and output-processing procedures, update frequency, token usage, and failure handling are included in Supplementary Section B.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
User message:
You are a conservative threshold advisor for
DexSafeDagger robot distillation.
Do not decide intervention directly. Recommend
threshold displacements for the existing gates.
Task: Propose displacements for the
teacher-student disagreement and predicted-success
thresholds used in dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys:
delta_l2_threshold, delta_success_threshold,
confidence, reason.
Keep changes conservative. Prefer small threshold
shifts unless the recent window shows clear over-
or under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min},
{success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
```

```
Recent rollout statistics:
{
 "frame": {frame}, "intervention_rate": {beta},
"step": {step},
 "sample_count": {N}, "l2_threshold":
{tau_delta},
 "success_threshold": {tau_Q},
"unsafe_episode_rate": {r_unsafe},
 "l2": {"min": {v}, "max": {v}, "mean": {v},
      "p10": {v}, "p50": {v}, "p90": {v}},
 "success": {"min": {v}, "max": {v}, "mean":
{v},
      "p10": {v}, "p50": {v}, "p90": {v}},
 "unsafe_reason_prop": { "hand_too_far": {v},
      "harmful_collision": {v},
"object_out_of_bound": {v},
      "palm_flipped": {v}},
 "visual_buffer_size": {B}, "visual_samples_attached":
{M},
 "visual_samples": [{ "env_id": {v}, "frame":
{v},
      "image_attached": true, "image_key":
{v}, "source": {high_l2 | low_success |
unsafe_triggered},
      "l2_threshold": {v}, "success_threshold":
{v},
      "teacher_student_l2": {v},
 "predictor_success": {v},
      "object_id": {v}, "object_name": {v},
"object_env_count": {v},
      "step": {v}, "unsafe": {true | false}}, ...]

}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata: {the corresponding
record above, excluding "image_attached"}
<IMAGE {i}: JPEG data URL, detail="high">
```

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

The requested machine-readable output is a top-level JSON object with the following schema:

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  "delta_l2_threshold": <number>, "delta_success_threshold": <number>,
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The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

Comment 6: ***Real-world unseen-object performance is weak: lift success falls to 60–66.7%, while unsafe episodes reach 40%. Claims of sufficient deployment robustness should be moderated.***

[Response:] We will remove the claim that the current results establish sufficient robustness for deployment.

Comment 7: ***Only fifteen real-world episodes per reported object produce coarse estimates. Increase repetitions and report confidence intervals and statistical comparisons across all 22 objects.***

[Response:] We have doubled the number of real-world evaluation episodes from 15 to 30 per object. We have also reported the complete results for all 22 tested objects in Supplementary Section C.3. Supplementary Table V reports the pooled means and Wilson 95% confidence intervals for the 14 seen and eight unseen objects.

Table 8: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 9: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

Comment 8: ***Safety-event thresholds for harmful collision, object escape, palm flipping, and workspace violation require physical justification. Include contact-force measurements and severity-weighted safety metrics.***

[Response:] We have provided the exact definitions of all unsafe events in the termination-conditions discussion in Supplementary Section A.2. When multiple unsafe conditions occur simultaneously during training, the event label is assigned according to a clear severity-based priority: harmful collision, object escape, palm flipped, and hand too far, in descending order.

Termination Conditions. We define four unsafe conditions that serve both as termination criteria during teacher-policy training and as unsafe-event signals during student-policy distillation. We additionally terminate an episode when it reaches the maximum time horizon; however, a timeout is not classified as an unsafe event. All five termination conditions are checked at 60 Hz in the environment frame.

(a) *Harmful collision:* We monitor all arm–table contacts and trigger this termination when the palm or any fingertip is less than 1 cm above the tabletop, or when any arm link contacts the table. An arm–table contact is registered when the ℓ_1 norm of the filtered contact-force vector exceeds 0.1 N.

(b) *Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.

(c) *Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90°, at which point the nominal grasp geometry is inverted.

(d) *Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.

(e) *Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

This order reflects decreasing immediacy of harm and loss of control: direct arm–table contact poses the clearest hardware risk, object escape represents loss of containment, palm inversion and workspace departure

are geometric failures of decreasing immediate consequence, and timeout is last because it is not classified as unsafe.

Comment 9: ***Clarify whether VLM adaptation occurs only during simulation distillation or during physical deployment. If absent at deployment, the role of scene-level VLM reasoning should not be overstated.***

[Response:] The VLM is queried only during simulation-based online distillation. During physical deployment, the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. We will state this distinction in the abstract, method, experimental setup, and limitations, and will describe the VLM as a training-time adaptation mechanism rather than a deployment-time safety monitor.

Comment 10: ***Compare training samples, teacher queries, interventions, computation, and final success across Vanilla DAgger, SafeDAgger, critic-based intervention, and VLM-guided intervention using multiple random seeds.***

[Response:] We repeat the complete distillation process for Vanilla DAgger (b), Vanilla SafeDAgger (c), critic-based intervention (e), and VLM-guided intervention (f) using five independent random seeds, $\{40, 41, 42, 43, 44\}$. All variants use the same number of training samples, and the teacher-intervention rates exhibit only minor variation across seeds. The final lift-success and unsafe-episode rates are reported as the mean $\pm$ standard deviation across seeds in Supplementary Section C.2.

Table 10: Comparison of Vanilla DAgger, Vanilla SafeDAgger, critic-based intervention, and VLM-guided intervention across five random seeds $\{40, 41, 42, 43, 44\}$. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Vanilla DAgger (b)	Vanilla SafeDAgger (c)	Critic-based intervention (e)	VLM-guided intervention (f)
Lift Success (%) $\uparrow$	87.1 ± 0.99	86.3 ± 0.92	86.8 ± 0.74	87.5 ± 0.98
Unsafe Rate (%) $\downarrow$	17.4 ± 0.52	16.7 ± 0.48	11.9 ± 0.58	9.3 ± 0.39

We thank the Editor and the reviewers again for their thoughtful feedback. The requested revisions will help us present the method's contribution more clearly, support its empirical claims more rigorously, and state its limitations more precisely.

Supplementary Material

Journal Title: IEEE Robotics and Automation Letters
Reference Number: 26-3841v1
Title of Paper: VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
Authors: Anonymous Authors

A RL Teacher Training and Safety Details

A.1 Asymmetric Teacher Observations

Table 1 details the asymmetric actor–critic observation spaces used during teacher-policy training. The critic receives a 133-dimensional privileged simulator state, whereas the actor receives an 87-dimensional observation consisting only of signals available at deployment. We inject noise into the actor observations during training to improve robustness and reduce the sim-to-real gap. The palm and five fingertips constitute the six hand links. A "twist" is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to the value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	**87**	**133**	

"–" denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

A.2 Termination Conditions

We define four unsafe conditions that serve both as termination criteria during teacher-policy training and as unsafe-event signals during student-policy distillation. We additionally terminate an episode when it reaches the maximum time horizon; however, a timeout is not classified as an unsafe event. All five termination conditions are checked at 60 Hz in the environment frame.

(a) *Harmful collision:* We monitor all arm–table contacts and trigger this termination when the palm or any fingertip is less than 1 cm above the tabletop, or when any arm link contacts the table. An arm–table contact is registered when the ℓ_1 norm of the filtered contact-force vector exceeds 0.1 N.

(b) *Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.

(c) *Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90°, at which point the nominal grasp geometry is inverted.

(d) *Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.

(e) *Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

B VLM Prompt and Inference Details

B.1 Visual-Input Storage and Example

Here, we provide representative examples of the selected visual inputs in Fig. 1. These images illustrate observations associated with the two intervention sources considered by our framework: high teacher–student disagreement and low predicted task success. The algorithm block that follows presents the semantic prompt, which combines these visual samples with recent rollout statistics and instructs the VLM to recommend conservative updates to the corresponding intervention thresholds.



(a) (b) (c) (d)

Figure 1: Representative visual inputs selected according to the two intervention signals. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
User message:
You are a conservative threshold advisor for
DexSafeDagger robot distillation.
Do not decide intervention directly.  Recommend threshold
displacements for the existing gates.
Task:  Propose displacements for the teacher-student
disagreement and predicted-success thresholds used in
dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys:  delta_l2_threshold,
delta_success_threshold, confidence, reason.
Keep changes conservative.  Prefer small threshold
shifts unless the recent window shows clear over- or
under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min}, {success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
```

```
Recent rollout statistics:
{
 "frame":  {frame}, "intervention_rate":  {beta}, "step":
{step},
 "sample_count":  {N}, "l2_threshold":  {tau_delta},
 "success_threshold":  {tau_Q}, "unsafe_episode_rate":
{r_unsafe},
 "l2":  {"min":  {v}, "max":  {v}, "mean":  {v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "success":  {"min":  {v}, "max":  {v}, "mean":  {v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "unsafe_reason_prop":  { "hand_too_far":  {v},
     "harmful_collision":  {v}, "object_out_of_bound":
{v},
     "palm_flipped":  {v}},
 "visual_buffer_size":  {B}, "visual_samples_attached":
{M},
 "visual_samples":  [{ "env_id":  {v}, "frame":  {v},
     "image_attached":  true, "image_key":  {v},
"source":  {high_l2 | low_success | unsafe_triggered},

     "l2_threshold":  {v}, "success_threshold":  {v},
     "teacher_student_l2":  {v}, "predictor_success":
{v},
     "object_id":  {v}, "object_name":  {v},
"object_env_count":  {v},
     "step":  {v}, "unsafe":  {true | false}}, ...]
}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata:  {the corresponding record
above, excluding "image_attached"}
<IMAGE {i}:  JPEG data URL, detail="high">
```

B.2 Model and Input Configuration

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

B.3 Output Processing and Threshold Updates

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "delta_l2_threshold": <number>, "delta_success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\Big(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\Big), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\mathrm{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

C Additional Experimental Results

C.1 Bellman Critic Evaluation

We use the resulting student and fixed-threshold Bellman critic from ablation (e) to evaluate the critic's performance. Both networks are frozen during evaluation, and the critic runs in parallel as a passive evaluator. We evaluate 200 episodes for each of the 40 training objects in simulation. Table 2 reports the observed lift-success and unsafe-episode rates together with the task-outcome prediction accuracy, the unsafe-episode detection rate, and its corresponding false-negative rate.

The lift-success rate and unsafe-episode rate characterize the student policy's ability to accomplish the task and the frequency of unsafe outcomes, respectively. Task-outcome prediction accuracy is the percentage of episodes for which the critic correctly predicts the task outcome. The unsafe-episode detection rate is the percentage of unsafely terminated episodes that are flagged by the critic before termination. The unsafe false-negative rate is the complementary percentage of unsafe episodes that the critic does not flag before termination.

Table 2: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

C.2 Ablations Across Random Seeds

We independently train four intervention variants—Vanilla DAgger (b), Vanilla SafeDAgger (c), critic-based intervention (e), and VLM-guided intervention (f)—using five random seeds, $\{40, 41, 42, 43, 44\}$. For each variant and seed, we freeze the resulting student and evaluate it for 200 episodes on each of the 40 training objects in simulation, yielding 8,000 evaluation episodes per seed. Because all variants use the same number of training samples and the teacher-intervention rates exhibit only minor variation across seeds, Table 3 reports only the final lift-success and unsafe-episode rates as the mean $\pm$ standard deviation across the five seeds.

Table 3: Comparison of Vanilla DAgger, Vanilla SafeDAgger, critic-based intervention, and VLM-guided intervention across five random seeds $\{40, 41, 42, 43, 44\}$. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Vanilla DAgger (b)	Vanilla SafeDAgger (c)	Critic-based intervention (e)	VLM-guided intervention (f)
Lift Success (%) $\uparrow$	87.1 ± 0.99	86.3 ± 0.92	86.8 ± 0.74	87.5 ± 0.98
Unsafe Rate (%) $\downarrow$	17.4 ± 0.52	16.7 ± 0.48	11.9 ± 0.58	9.3 ± 0.39

C.3 Complete Real-World Evaluation and Demonstrations



Figure 2: **Sixteen-object demonstration inventory:** (1) Horse, (2) Teapot, (3) Sphinx, (4) Dragon, (5) Homer bust, (6) Ogre, (7) Cheburashka, (8) Lucy statue, (9) Beethoven bust, (10) Igea bust, (11) Dog pair, (12) Cow statue, (13) Buddha statue, (14) Blub character, (15) Monkey head, and (16) Dinosaur.

We report results for the remaining 16 real-world test objects that are not shown in the manuscript's six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes.

Table 4: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 5 summarizes performance across all 22 objects. The mean rates reproduce the pooled results in Main Paper Table IV; the 95% confidence intervals account for the 420 seen-object episodes and 240 unseen-object episodes.

Table 5: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

C.4 Physical Baseline Comparison

We deploy the frozen RGB student policies resulting from ablations (e) and (f) on the real robot. Ablation (e) uses our RL teacher-training recipe with the stationary DexSafeDagger intervention policy, whereas ablation (f) uses the same teacher recipe with VLM-guided threshold adaptation. We evaluate both policies using the same objects, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates separately for the 14 seen and 8 unseen objects in Table 6.

Table 6: Overall real-world comparison of the frozen student policies resulting from ablations (e) and (f). Rates are reported separately for the 14 seen and 8 unseen objects.

Metric (%)	Ablation (e)		Ablation (f)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

The comparison supports our claim that the student trained with VLM-augmented intervention achieves a comparable real-world task-success rate while exhibiting a lower unsafe-event rate for both seen and unseen objects.

C.5 Implementation Details of Ablation (h)

Ablation (h) replaces the VLM tuner in ablation (f) with a deterministic performance-triggered threshold curriculum inspired by the budget-aware intervention principle of ThriftyDAgger. The teacher, student, Bellman critic, intervention gate, admissible threshold range, rollout budget, and update schedule are otherwise identical to those in ablation (f); the curriculum uses neither language nor visual inputs.

At scheduled update k, let $\mathscr{W}_k$ denote the most recent rollout window. We compute its lift-success and unsafe-episode rates as

$$\bar{s}_k = \frac{1}{|\mathscr{W}_k|} \sum_{\tau \in \mathscr{W}_k} \mathbb{I}_{\text{succ}}(\tau), \qquad \bar{u}_k = \frac{1}{|\mathscr{W}_k|} \sum_{\tau \in \mathscr{W}_k} \mathbb{I}_{\text{unsafe}}(\tau). \tag{1}$$

The curriculum contains five ordered operating points, indexed from the most conservative ($j = 0$) to the most autonomous ($j = 4$):

j	0	1	2	3	4
$c_\delta^{(j)}$	0.50	0.75	1.00	1.25	1.50
$c_Q^{(j)}$	1.50	1.25	1.00	0.75	0.50

(2)

At level j_k, the intervention thresholds are

$$\tau_\delta^{(k)} = c_\delta^{(j_k)} \tau_\delta^{(0)}, \qquad \tau_Q^{(k)} = c_Q^{(j_k)} \tau_Q^{(0)}. \tag{3}$$

Thus, moving toward a smaller level lowers the disagreement threshold and raises the minimum acceptable predicted-success threshold, producing more teacher intervention. Moving toward a larger level has the opposite effect and grants the student more autonomy. The resulting gate remains an OR rule: intervention occurs when either $\|a_t^S - a_t^T\|_2 > \tau_\delta^{(k)}$ or $\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)}$.
Let $u_{\max}$ be a validation-selected unsafe-rate limit and $s_{\min}$ a validation-selected competence threshold. Define $A_k = \mathbb{I}[\bar{u}_k \leq u_{\max} \wedge \bar{s}_k \geq s_{\min}]$. Starting from the nominal operating point $j_0 = 2$, the curriculum update is

$$j_{k+1} = \begin{cases} \max(j_k - 1, 0), & \bar{u}_k > u_{\max}, \\ \min(j_k + 1, 4), & A_k = 1 \ \wedge \ A_{k-1} = 1, \\ j_k, & \text{otherwise.} \end{cases} \tag{4}$$

Consequently, one unsafe window immediately moves the gate one level toward more teacher intervention, whereas two consecutive windows satisfying both safety and task-performance criteria are required before increasing student autonomy. This asymmetric update prevents rapid oscillation. We apply the first curriculum update after 2,000 distillation episodes and subsequent updates every 10,000 episodes, matching the VLM update schedule in ablation (f).

VLM-DexSafeDagger: VLM-Guided Intervention Policy ~~for Safety-Critical Visuomotor Skill Distillation~~ Towards Safer Visuomotor Skill Distillation

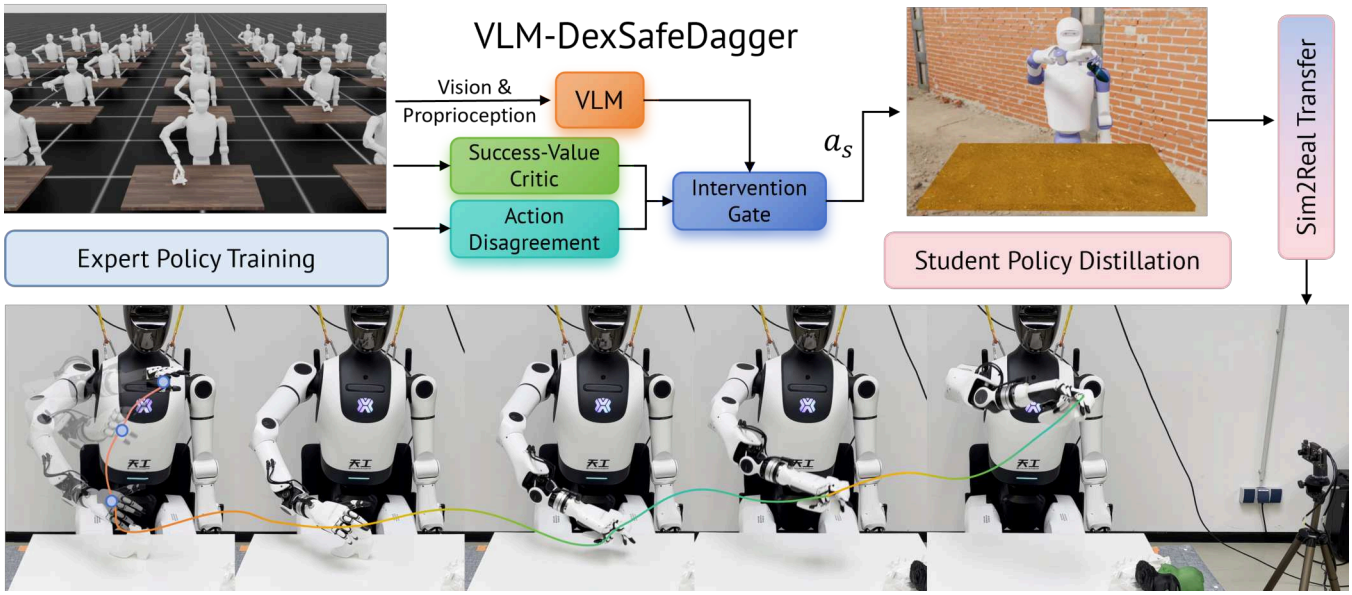


Fig. 1: **VLM-DexSafeDagger overview.** We propose a teacher-intervened online distillation pipeline that condenses teacher skills into a deployable RGB-based student policy. The unsafe exploration rate is substantially suppressed in the early training phase via a proposed VLM-guided intervention policy that combines teacher–student action disagreement with a Bellman success-value critic trained on safe-lift-success labels, while a vision-language model maintains an evolving, safety-aware understanding of the scene and periodically adjusts the intervention policy during online distillation.

***Abstract*— Learning dexterous manipulation is challenging due to high-dimensional action spaces and complex contact dynamics. In addition, safety-unaware exploration during learning can lead to hardware damage and unstable or even hazardous contact events, posing a significant barrier to real-world deployment. To address this challenge, we propose VLM-DexSafeDagger, a safety-aware online skill-distillation framework that transfers reinforcement learning (RL) teacher policies to a vision-based student policy. VLM-DexSafeDagger combines an ℓ_2 teacher–student disagreement gate with a learned Bellman success-value critic that estimates the prospective safe-lift-success value of the current student state–action pair. A vision-language model periodically adjusts the intervention policy parameterized by disagreement limit and acceptable predicted success rate. This design enables safer exploration during online interactive imitation learning using high-level visual context. Experiments on grasp-and-lift tasks show that VLM-DexSafeDagger significantly improves training safety, reducing unsafe episodes by up to 80% during the training period compared with vanilla DAgger-based distillation. Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates ~~sufficient robustness for real-world robotic deployment~~successful transfer from simulation to the real robot.**

I. INTRODUCTION

Dexterous robotic hands typically possess a large number of degrees of freedom (DoFs) and exhibit complex contact dynamics, enabling them to perform fine-grained and diverse manipulation behaviors~~, such as multi-finger grasping [1], in-hand reorientation [2], [3], and precise control [4]. However, the high-dimensional action space and intricate interaction dynamics substantially increase the difficulty of control and policy learning~~ [1]–[4]. In recent years, reinforcement learning (RL)-based approaches [3], [5]–[7] have ~~achieved remarkable progress in dexterous grasping and manipulation tasks. Leveraging large-scale parallel simulation and high-fidelity physics engines, RL enables the automatic learning of complex control policies in simulated environments without the need for manually designed controllers. These methods have demonstrated strong generalization capabilities across multi-object and multi-task scenarios~~made significant progress in addressing the complex control problems posed by high-dimensional action spaces and intricate contact dynamics in dexterous

manipulation tasks.

Despite these advances, existing RL frameworks primarily optimize for task success rates and rarely incorporate explicit modeling or constraints related to ~~safety-critical behaviors~~safety. During training, common failure modes~~include~~ , including severe collisions between the hand and the environment, unintended object ejection due to excessive contact forces, and loss of stable hand posture [8], [9]~~. Such unsafe behaviors~~ , not only reduce training efficiency but also introduce potential risks during real-world deployment. Therefore, how to reduce unsafe exploration while maintaining learning efficiency remains a critical challenge in dexterous manipulation.

To ~~mitigate the performance gapbetween simulation and real-world deployment~~bridge the sim-to-real gap, skill distillation ~~has become a widely adopted paradigm in recent years. The core idea is~~ is widely used to transfer the behavior of expert policies trained with privileged information to ~~first train a high-performing expert policy in simulation using privileged state information, and then distill its decision-making behavior into a student policy that relies solely on visual observations or partially observable inputs. This process enables modality transfer from privileged state information to perception-based inputs, facilitating efficient real-world deployment. Skill distillation reduces reliance on accurate state estimation and allows knowledge integration from multiple teacher policies , thereby substantially improving execution stability and generalization performance in real-world environments. Consequently, it has gained increasing attention in dexterous grasping and complex manipulation tasks~~student policies capable of operating on real robots. However, existing distillation methods [10], [11] typically rely on behavior cloning or the standard DAgger framework for data aggregation. Their primary objective is to mitigate performance degradation caused by distribution shift, rather than explicitly enforcing safety constraints during student policy learning. During distillation, the student policy may deviate from the teacher's state–action distribution and enter regions that are either insufficiently covered by the teacher or inherently high-risk, leading to unsafe behaviors such as severe collisions or unstable hand configurations. Therefore, introducing explicit safety mechanisms into the distillation process—so that high-risk exploration can be avoided without sacrificing learning efficiency—remains an important and largely underexplored problem. ~~Existing~~ Typical safety-aware distillation approaches, such as SafeDAgger [12], have primarily focused on autonomous driving scenarios and have not been systematically studied in ~~the context of~~ contact-rich tasks such as dexterous manipulation.

The contributions of this work are summarized as follows:

- We propose VLM-DexSafeDagger, a VLM-guided, safety-sensitive framework for distilling RL-acquired teacher policies into a single RGB-based student policy. To the best of our knowledge, this is the first attempt in dexterous manipulation to incorporate a mixed intervention policy combining teacher–student action disagreement with a Bellman success-value critic, while the VLM reasons about scene-level safety from rollout statistics and representative visual observations to regulate the intervention gate.
- We propose a safety-aware RL reward design that explicitly accounts for risky events in arm–hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training.
- We empirically demonstrate that VLM-DexSafeDagger, which combines disagreement, success-value prediction, and VLM-guided intervention policy adaptation, substantially reduces unsafe exploration compared with Vanilla DAgger, achieving an approximately **80%** relative reduction in the unsafe-episode rate during early training. It also reduces the average unsafe-episode rate by **7.5 percentage points** relative to disagreement-only SafeDAgger [12]. Ablation results further demonstrate the effectiveness of VLM-guided adaptation across different visual contexts and unsafe scenarios: VLM-DexSafeDagger consistently achieves lower unsafe-episode rates than DexSafeDagger, with an average reduction of **4 percentage points**.

II. Related Work

Skill Distillation. Skill distillation transfers behavior from a high-capacity teacher to a compact student and has been widely adopted in both computer vision and robotics. In robotics, the teacher can leverage privileged state and simulator access during training, while the student is distilled to operate from on-board observations (e.g., RGB). DexTrAH-RGB [11] distills a privileged RL teacher into an RGB-based policy via DAgger-based imitation learning, while UniGraspTransformer [13] distills offline trajectories from object-specific teachers into a unified Transformer policy.

Adaptive Intervention in Interactive Imitation Learning. Many imitation learning methods ~~operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution~~use fixed offline datasets for supervision [14], [15]. In sequential decision-making, however, ~~distributional~~ distribution shift induces covariate shift and compounding errors. Interactive imitation learning ~~[16] mitigates this issue by collecting supervision online~~ [16] mitigates these issues by collecting online supervision along trajectories visited by the learner.

~~DAgger~~DAgger [10] addresses covariate shift by iteratively ~~rolling out the learner and relabeling visited~~ relabeling learner-visited states with an expert, ~~thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over~~ while SafeDAgger [12] adds teacher intervention in high-risk states to reduce unsafe exploration. ~~Recent variants [17], [18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision~~

Later methods introduce adaptive intervention mechanisms based on uncertainty-driven rules [19], human gating [20]

, or learned intervention criteria [21]. However, they have primarily been evaluated in relatively simple control or autonomous-driving settings. Fully autonomous expert relabeling during online distillation remains rare in robotic manipulation, especially for dexterous, contact-rich tasks.

Vision-Language-Guided Robot Learning. Visual and semantic information is widely recognized as a strong cue for embodied agents to interpret task specifications and reason about unstructured environments [22], [23]. One line of work uses VLMs to generate demonstrations or supervisory signals for offline policy construction, such as Manipulate-Anything [24]and AHA [25]. Complementary approaches, including RACER [26] and FOREWARN [27], employ VLMs to guide risk prediction, provide recovery guidance for unsafe situations, or govern action selection during online policy learning or real-time execution.

III. Problem Statement and Methods

A. Problem Formulation

We consider online interactive imitation learning with teacher intervention in a partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, r, \gamma)$, with privileged teacher state $s_t \in \mathcal{S}$, deployable student observation $o_t \in \mathcal{O}$ (RGB), action $a_t \in \mathcal{A}$, observation model $o_t \sim \Omega(\cdot \mid s_t)$, and dynamics $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. During training, we assume access to a set of privileged teacher policies $\{\pi_{T_i}\}_{i=1}^{K}$, whereas at deployment the student policy π_S must act solely from o_t. Our goal is to distill the collective teacher behavior into an RGB-based student that achieves high task success while minimizing unsafe exploration during training.

The induced rollout distribution $p_{\pi_\mathcal{G}}$ is defined by the gated rollout policy $\pi_\mathcal{G}$, which switches between the teacher and student policies according to the intervention gate. Its learning objective and intervention gate are

$$\min_{\substack{\pi_S \in \Pi_S \\ \mathcal{G} \in \mathfrak{G}}} \Bigg[\underbrace{-\mathbb{E}_{\tau \sim p_{\pi_\mathcal{G}}}[\mathbb{I}_{\text{succ}}(\tau)]}_{\text{maximize success}} + \lambda \underbrace{\mathbb{E}_{\tau \sim p_{\pi_\mathcal{G}}}\Big[\sum_{t=0}^{T-1} \mathbb{I}_{\text{unsafe}}(s_t, a_t)\Big]}_{\text{minimize unsafe}} \Bigg]$$

$$\begin{aligned} \text{s.t.} \quad & a_t = g_t a_t^T + (1 - g_t) a_t^S, \\ & g_t = \mathcal{G}(s_t, o_t; \boldsymbol{\tau}^{(k)}) \in \{0, 1\}, \\ & s_{t+1} \sim P(\cdot \mid s_t, a_t), \qquad o_t \sim \Omega(\cdot \mid s_t). \end{aligned} \tag{1}$$

where $g_t = 1$ indicates teacher takeover and $g_t = 0$ denotes student control. The intervention policy is parameterized by $\boldsymbol{\tau}^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, which governs teacher takeover decisions in $\mathcal{G}$. Periodically, a VLM proposes a displacement $\Delta\boldsymbol{\tau}^{(k)}$, from which the policy is updated to adjust to distributional shift.

We formalize task success via a trajectory-level predicate $\mathbb{I}_{\text{succ}}(\tau) \in \{0, 1\}$, which is 1 if the object is grasped, lifted, and held above a height threshold for a consecutive duration. Safety is governed by early termination conditions: letting $\mathbb{I}_{\text{unsafe}}(s_t, a_t) \in \{0, 1\}$ indicate whether any unsafe event (e.g., harmful collision, object escape, palm flip, hand-too-far) occurs at time t, the trajectory is safe if $\prod_{t=0}^{T-1} \big(1 - \mathbb{I}_{\text{unsafe}}(s_t)\big) = 1$.

Algorithm 1 VLM-DexSafeDagger

Train the privileged teacher policy π_T via RL
Collect a warm-start dataset $\mathcal{D}_0$ under teacher control
Pretrain the Bellman critic $\hat{Q}_{\text{succ}}$ on $\mathcal{D}_0$
Initialize the intervention policy parameters $\boldsymbol{\tau}^{(0)} = (\tau_\delta^{(0)}, \tau_Q^{(0)})$
Initialize the visual buffer $\mathcal{B}_V$
for iteration $k = 1 \dots N$ **do**
  Observe the teacher state s_t and student observation o_t
  Compute the actions $a_t^T \sim \pi_T(\cdot|s_t)$ and $a_t^S \sim \pi_S(\cdot|o_t)$
  Compute the action disagreement $\delta_t = \|a_t^S - a_t^T\|_2$
  Predict the safe-lift-success value $\hat{Q}_{\text{succ}}(s_t, a_t^S)$
  Evaluate the intervention gate
    $g_t \leftarrow \mathbb{I}[\delta_t > \tau_\delta \ \vee \ \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q]$
  Execute the gated action $a_t = g_t a_t^T + (1 - g_t) a_t^S$
  Store the transition in the critic replay buffer $\mathcal{D}$
  Update $\hat{Q}_{\text{succ}}$; if $g_t = 1$, update π_S
  Add representative visual states to $\mathcal{B}_V$
  if an intervention policy update is due **then**
    Summarize rollout context $\mathcal{C}_k$ and sample images $\mathcal{V}_k$
    Query the VLM for the shift $\Delta\boldsymbol{\tau}^{(k)}$ given $(\mathcal{C}_k, \mathcal{V}_k)$
    Form the candidate $\tilde{\boldsymbol{\tau}}^{(k)} = \boldsymbol{\tau}^{(k)} + \Delta\boldsymbol{\tau}^{(k)}$
    Clip and smooth $\tilde{\boldsymbol{\tau}}^{(k)}$ to obtain $\boldsymbol{\tau}^{(k+1)}$
  end if
end for

B. Overview of the Distillation Pipeline

Fig. 2 summarizes VLM-DexSafeDagger, our teacher–student distillation pipeline, which consists of five core components: (i) privileged teacher policy training in simulation (Sec. C), (ii) Bellman success-value critic learning to estimate safe lift success under student actions (Sec. D), (iii) deterministic teacher intervention (Sec. E), (iv) VLM-guided intervention policy adaptation (Sec. F), and (v) online student skill distillation (Sec. G).

The privileged teacher is first trained in simulation, and its rollouts provide an offline warm start for the Bellman critic. The critic estimates prospective lift success under the student action and continues to be updated from a ring replay buffer.

During online distillation, the teacher intervenes when either teacher–student action disagreement is high or the critic predicts low success. The student is then optimized online from teacher labels collected under the resulting rollout policy $\pi_\mathcal{G}$. A VLM is periodically queried to propose updates to the intervention policy's operating point. The resulting candidate operating point is clipped and smoothed before use. Pseudo-code is provided in Alg. 1.

C. Privileged Teacher Policy Training

Teacher policy learning is formulated as a reinforcement learning problem with transition tuples $\langle s_t, a_t, r_t, s_{t+1} \rangle$, where s_t is the privileged state, a_t is the action, r_t is the reward, and s_{t+1} is the next state. The objective is to maximize the expected discounted return,

$$\pi_T^* = \arg\max_{\pi_T} \ \mathbb{E}_{\pi_T}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right]. \tag{2}$$

We use a shaped reward $r_t = \sum_{k=1}^{K_r} r_{t,k}$ to encourage stable grasp acquisition, object lifting, and goal reaching (Table I).

The teacher is trained in simulation using an asymmetric

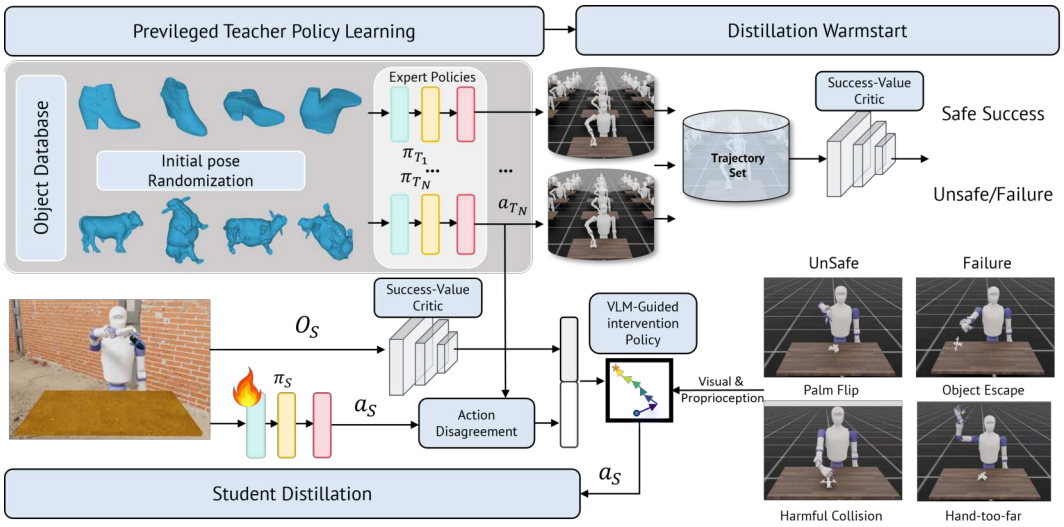


Fig. 2: Overview of the VLM-DexSafeDagger teacher–student learning pipeline. A privileged reinforcement learning teacher is trained in simulation and subsequently distilled into an RGB-based visuomotor policy for real-world deployment. A vision-language model periodically proposes a displacement of the intervention policy's operating point; deterministic clipping and smoothing constrain the resulting policy update.

TABLE I: Reward components used for teacher policy training.

Term	Name	Purpose
r_{dist}	Hand distance	Guides the hand toward the object.
r_{curl}	Finger curl	Encourages a grasp-ready finger posture.
r_{align}	Palm alignment	Aligns the palm to a palm-down pose.
r_{contact}	Contact bonus	Rewards stable thumb–finger contact.
r_{near}	Stay-close	Keeps the hand near the object after reaching.
r_{goal}	Goal tracking	Guides the grasped object toward the goal.
r_{lift}	Lift	Rewards lifting after stable contact.
r_{action}	Action smoothness	Penalizes abrupt action changes.

actor–critic setup. Details of the actor and critic observation spaces are provided in the Appendix. The actor receives noisy and partial proprioceptive observations that approximate realistic sensing, while the critic is provided with the privileged full state of both the robot and object, available only in simulation.

Episodes terminate at

$$T = \min\{t \,|\, \mathbb{I}_{\text{unsafe}}(s_t) = 1 \;\text{ or }\; t \geq T_{\max}\}, \tag{3}$$

where $T_{\max} = 10\,\text{s}$ is the timeout horizon. Early termination conditions include palm flip, harmful collision, object escape, and hand-too-far (see Sec. A), and are treated as unsafe events for both teacher training and subsequent intervention decisions.

D. Bellman Success-Value Critic

ℓ_2 action disagreement alone is insufficient as an intervention signal and may miss small action deviations that could cause unstable grasps, collisions, or object escape. We therefore learn a Bellman critic, trained with safe-lift-success labels, that estimates the probability of safe lift success for the current student state–action pair and provides a prospective signal complementary to action disagreement.

The Bellman critic evaluates a multimodal state–action pair (s_t, a_t^S), where s_t combines robot proprioception and embeddings from a visual encoder. The encoder is trained end-to-end during skill distillation and regularized by the auxiliary object-position prediction loss in Sec. G. The twin critics Q_1 and Q_2 are ReLU MLPs with hidden dimensions $[256, 128]$. Each critic's logit $z_i(s,a)$ is mapped to a success probability by $Q_i(s,a) = \sigma(z_i(s,a)/T)$, with calibration temperature $T = 4$. We use the conservative estimate $\hat{Q}_{\text{succ}}(s,a) = \min_{i\in\{1,2\}} Q_i(s,a)$; lower values indicate higher prospective risk.

The critics are pretrained on teacher transitions and updated online from a ring replay buffer. The label $u_t \in \{0,1\}$ indicates safe lift success: it is 1 only when the lift criterion is satisfied without an unsafe event. Training uses the SARSA-style target

$$y_t = \text{clip}\Big(u_t + \gamma(1-d_t^{\text{eff}}) \min_{i\in\{1,2\}} Q_i^{\text{targ}}(s_{t+1}, a_{t+1}),\, 0,\, 1\Big), \tag{4}$$

where d_t is the termination flag, h_t indicates whether a_{t+1} is valid, $d_t^{\text{eff}} = \max(d_t, 1-h_t)$, and Q_i^{targ} are slowly updated target networks. Both critics minimize the mean-squared error (MSE) between their predictions and y_t; the minimum across critics reduces overestimation.

E. Intervention Policy

The intervention policy uses a deterministic two-channel gate. The first channel measures teacher–student disagreement, $\delta_t := \|a_t^S - a_t^T\|_2$, where $a_t^T \sim \pi_T(\cdot \mid s_t)$ and $a_t^S \sim \pi_S(\cdot \mid o_t)$. The second channel evaluates prospective task success through $\hat{Q}_{\text{succ}}(s_t, a_t^S)$. The intervention policy is parameterized by $\boldsymbol{\tau}^{(k)} = (\tau_\delta^{(k)}, \tau_Q^{(k)})$, where $\tau_\delta^{(k)}$ is the maximum admissible disagreement and $\tau_Q^{(k)}$ is the minimum acceptable predicted success.

These parameters define the deterministic intervention gate:

$$g_t = \mathbb{I}\Big[\hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q^{(k)} \ \vee \ \delta_t > \tau_\delta^{(k)}\Big]. \tag{5}$$

Thus, either large disagreement or low predicted success triggers teacher takeover. When the two signals conflict, the conservative OR rule applies: low teacher–student disagreement does not suppress intervention if the critic predicts low student success, and a favorable critic prediction does not suppress intervention if the action disagreement is large. The student acts only when neither intervention condition is triggered. Teacher takeover is treated as empirically safer rather than as an absolute safety guarantee, and the resulting safety performance remains bounded by the teacher policy's own performance ceiling. Decreasing τ_δ or increasing τ_Q makes the intervention policy more conservative; the opposite shift increases student autonomy.

We empirically initialize $\tau_\delta^{(0)}$ and $\tau_Q^{(0)}$ from the 90th percentile of warm-start disagreement and the 10th percentile of predicted success over successful teacher rollouts, respectively. These values define the initial operating point $\boldsymbol{\tau}^{(0)}$.

F. VLM-Guided Intervention Policy Adaptation

During online distillation, the student policy, visited-state distribution, and Bellman critic's calibration co-evolve, and the dominant failure modes may also change. A stationary intervention policy therefore cannot maintain a consistent safety–autonomy trade-off. We periodically query a VLM to guide the displacement of the intervention policy's operating point. At iteration k, the VLM receives rollout statistics $\mathcal{C}_k$, including disagreement (δ_t), predicted success ($\hat{Q}_{\text{succ}}(s_t, a_t^S)$), intervention rate ($\beta$), and unsafe-episode rate (UnsafeRateAvg). It also receives a representative image batch $\mathcal{V}_k$ sampled from the visual buffer $\mathcal{B}_V$ to cover high-disagreement, low-success, and unsafe states. From this context, it proposes an operating-point displacement $\Delta\boldsymbol{\tau}^{(k)} = (\Delta\tau_\delta^{(k)}, \Delta\tau_Q^{(k)})$, producing the candidate operating point $\tilde{\boldsymbol{\tau}}^{(k)} = \boldsymbol{\tau}^{(k)} + \Delta\boldsymbol{\tau}^{(k)}$.

Each candidate policy parameter is clipped around its initial value and exponentially smoothed:

$$\tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha \ \text{clip}\Big(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\Big), \quad x \in \{\delta, Q\}. \tag{6}$$

Here, α controls the adaptation rate, and $c_x^{\min}$ and $c_x^{\max}$ define the admissible calibration region around $\tau_x^{(0)}$.

G. Student Skill Distillation

Student learning is posed as supervised imitation from teacher-labeled transitions collected under the gated rollout policy $\pi_{\mathcal{G}}$. The student observes RGB images (two stereo views) together with robot proprioception.

We optimize a combined objective $\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{aux}}$, where the auxiliary loss predicts object position to regularize the visual encoder, $\mathcal{L}_{\text{aux}} = \|\hat{\mathbf{x}}_{\text{obj}} - \mathbf{x}_{\text{obj}}\|_2$.

For action imitation, we use a weighted L_2 loss, $\mathcal{L}_{\text{action}} = \|\mathbf{a}_S - \mathbf{a}_T\|_W^2 = (\mathbf{a}_S - \mathbf{a}_T)^\top W(\mathbf{a}_S - \mathbf{a}_T)$, with $W = \text{diag}(w_1, \ldots, w_d)$. We set $w_j = \sigma_{0,j}^{-2}$ using a reference scale $\sigma_{0,j}$ (e.g., the teacher action standard deviation on dimension j), yielding $\mathcal{L}_{\text{action}} = \sum_{j=1}^{d}(a_{S,j} - a_{T,j})^2/\sigma_{0,j}^2$.

Training proceeds in two stages. (i) *Warm-start:* we collect an offline dataset (e.g., 100k transitions) under full teacher control ($g_t \equiv 1$) to pretrain the Bellman critic. (ii) *Online distillation:* we perform interactive rollouts under $\pi_{\mathcal{G}}$ and query teacher labels on intervened steps. The student is updated online using the current-step teacher label only; no replay buffer or offline dataset is maintained for student updating. The design reduces the effect of stale supervision and stabilizes online imitation updates.

IV. Experiment

A. Evaluation Metrics

We designed the following metrics to evaluate both teacher and student policy performance. Let $\tau \in \{1, \ldots, N\}$ index evaluation episodes and $t \in \{0, \ldots, T_\tau\}$ index timesteps. Unless otherwise stated, we run the evaluation episodes with a horizon of 10 s.

a) Lift success (ever-lifted with hold).

Let h_t^τ be the object height above the table and H a lift threshold. An episode is considered successful if the object remains above H for at least N_{hold} consecutive steps. We define $\mathbb{I}_{\text{lift}}^\tau = \mathbb{I}\big[\exists\, t:\ \forall k \in \{0, \ldots, N_{\text{hold}} - 1\},\ h_{t+k}^\tau \geq H\big]$ and report: $\text{LiftSucc} = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_{\text{lift}}^\tau$.

b) Unsafe episode rate.

As stated above, an episode may terminate early due to one of the following conditions: palm flipped ($\mathbb{I}_{\text{flip}}^\tau$), harmful collision ($\mathbb{I}_{\text{coll}}^\tau$), object escape ($\mathbb{I}_{\text{obj}}^\tau$), or hand too far ($\mathbb{I}_{\text{hf}}^\tau$). Let $\mathcal{M} = \{\text{coll}, \text{obj}, \text{flip}, \text{hf}\}$ denote the set of termination modes, and let $\mathbb{I}_m^\tau \in \{0, 1\}$ indicate whether mode $m \in \mathcal{M}$ occurs in episode τ. An episode is unsafe if any termination mode is triggered: $\mathbb{I}_{\text{unsafe}}^\tau = \mathbb{I}\big[\bigvee_{m\in\mathcal{M}} \mathbb{I}_m^\tau = 1\big]$. We report: $\text{UnsafeRateAvg} = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_{\text{unsafe}}^\tau$.

c) Failure-mode decomposition.

Each episode is assigned a single terminal mode $m^\tau \in \mathcal{M}$ from the set of unsafe events (collision, object escape, palm flip, hand-too-far) as well as timeout/success where applicable. We define $\mathbb{I}_m^\tau = \mathbb{I}[m^\tau = m]$ and report $\text{FailureModeRate}(m) = \frac{1}{N}\sum_{\tau=1}^{N} \mathbb{I}_m^\tau, m \in \mathcal{M}$.

When multiple unsafe conditions are triggered in the same timestep, we record the first triggered termination condition (using a fixed priority order) as the episode's failure mode.

B. Simulation Experiment Setup

Simulation ~~experiments use~~ uses single-arm control ~~on~~ of a Tiangong 2 Pro humanoid ~~equipped with~~ with an InspireHand RH56DfX ~~; the action space is 13-D~~ and a 13-D action space (7 arm ~~DoF~~ + 6 hand DoF). The RGB student ~~additionally~~ receives two 320 × 240~~camera streams at 30 fps~~, 30-fps camera streams, encoded by a ~~Siamese~~ ResNet-18 ~~visual backbone with view fusion~~ and fused into a 128-D feature appended to the actor observation. Teacher training and student distillation ~~are conducted~~ run in Isaac Sim 5.0.0 / Isaac Lab 2.2.1 on ~~a single~~ one RTX 5090 GPU.

We ~~evaluate~~ train 40 ~~objects from the~~ object-specific teachers on Omniverse Asset Library ~~and train one~~

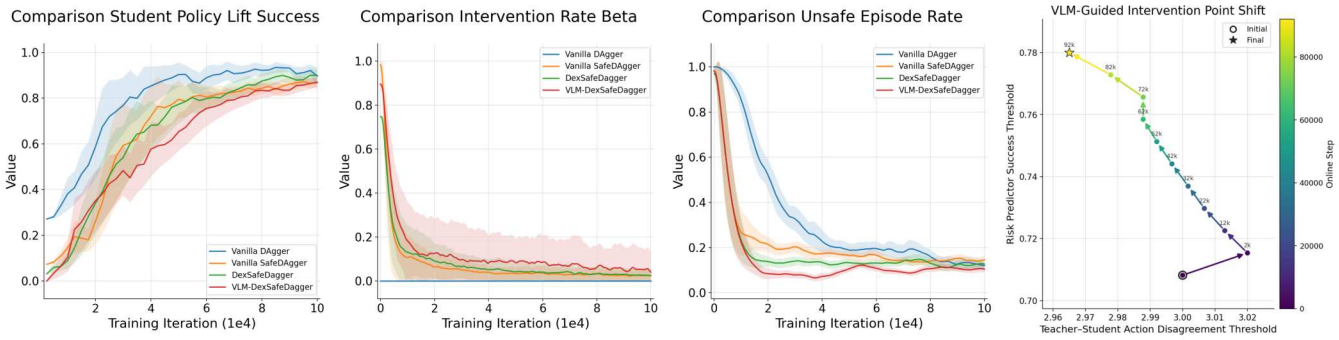


Fig. 3: **Ablation studies.** The first three panels compare four distillation variants in terms of student lift success, teacher intervention rate β, and unsafe episode rate, respectively, all plotted over training iterations. The rightmost panel shows the VLM-guided trajectory of the intervention operating point over online training. Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to vanilla DAgger (blue), vanilla SafeDAgger (orange), DexSafeDagger (green), and VLM-DexSafeDagger (red).

TABLE II: Teacher policy performance on four representative objects from the 40-object training set (200 evaluation episodes per object).

Metric (%)	WindCart	Boot	ToolBox	Cow	**Mean**
Lift Success	93.5	100.0	88.5	87.0	92.3
Unsafe Rate Avg	6.5	13.5	13.5	13.5	11.8
Harmful collision	0.0	0.0	6.5	0.0	1.6
Object escape	6.5	6.5	6.5	13.5	8.3
Palm flipped	0.0	6.5	0.0	0.0	1.6
Hand too far	0.0	0.0	0.0	0.0	0.0

~~object-specific teacher for each object. Each teacher is trained~~ objects, each for 12,000 episodes (3.5 h), totaling 480,000 ~~teacher-training~~ episodes and approximately 140 GPU-hours. ~~Across the 40 teachers, lift-success rates range from~~ Lift-success rates span 72–100% ~~,~~ and unsafe rates ~~range from~~ 5–30%; Table II ~~reports a representative four-object subset. During distillation, we randomize lighting intensity, direction, background textures, object material, and table appearance following DexTrAH-RGB-style domain randomization~~shows four representative objects. The joint student is distilled across all 40 objects for 100,000 episodes ~~in~~ (2.5 h), using DexTrAH-RGB-style randomization of lighting intensity and direction, background textures, object materials, and table appearance.

We use GPT-5.5 with high reasoning effort as the VLM. We issue the first VLM query after 2,000 episodes and subsequently query it every 10,000 distillation episodes, resulting in 10 VLM calls over the full 100,000-episode training run and approximately 70,000 tokens of total usage per run.

C. Ablation Studies

~~The~~ We perform the following ablation studies to validate the design choices in VLM-DexSafeDagger ~~. We distill the teacher policies trained during the RL phase into a compact, generalizable student policy. For RL training, we compare the standard DexTrAH-RGB pipeline with our customized variant. For skill distillation, we evaluate four methods—Vanilla DAgger, Vanilla SafeDAgger, , and —in terms of the evolution of lift-success, unsafe-episode, and intervention rate throughout training. The intervention rate is defined as the fraction of time steps at which the teacher takes over: $\beta = \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{I}[g_t = 1]$. We average this metric across evaluation episodes. As our primary safety metric, we report the unsafe-episode rate averaged across the four ablation objects, together with breakdowns by object and failure mode. The ablation settings are defined below.~~ and answer four questions: **(i) Does our customized RL recipe produce stronger teachers than the DexTrAH-RGB-style training recipe? (ii) Does the intervention mechanism in DexSafeDaggerenable safer exploration during policy distillation? (iii) Does augmenting action disagreement with success-value prediction improve the intervention signal? (iv) Is vision-language reasoning necessary for adapting the intervention boundary?**

(a) DexTrAH-RGB RL + Vanilla DAgger. The teacher is trained using the DexTrAH-RGB RL recipe ~~, and the student policy is trained using Vanilla DAgger. The student controls the rollout~~ (and then distilled into a student policy using supervision only, with no teacher intervention during rollouts ($\beta = 0$~~), while the parallel teacher policy labels each executed student action a_t^S with the corresponding teacher action a_t^T~~).

(b) Our RL + Vanilla DAgger. This setting matches (a)~~except that the teacher is trained with our customized RL recipe instead of~~ , except that our RL recipe replaces the DexTrAH-RGB ~~pipeline~~teacher-training recipe.

(c) Our RL + Vanilla SafeDAgger. ~~Using the same teacher as~~ The teacher training follows the recipe in (b), but we replace Vanilla DAgger with a disagreement-gated SafeDAgger baseline. The teacher intervenes only when $\|a_t^T - a_t^S\|_2$ exceeds a fixed threshold~~, and we train the student on teacher-labeled pairs (o_t^S, a_t^T)~~.

(d) DexTrAH-RGB RL + DexSafeDagger. ~~denotes SafeDAgger with disagreement and~~ We use the teacher-training recipe from (a) and the intervention baseline from (c), extending it with a Bellman success-

TABLE III: Performance of ablations (a)–(~~f~~h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5	72.7	26.5
(b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0	72.3	25.8
(e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(g)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(h)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

value critic under a stationary intervention policy. ~~This setting uses the DexTrAH-RGB-trained teacher (as in (a)). Intervention~~ Teacher intervention is triggered by either large teacher–student disagreement or low predicted success~~, and the student is trained from teacher-labeled pairs (o_t^S, a_t^T) collected under this gate~~.

(e) Our RL + DexSafeDagger. ~~This setting uses the setup defined~~ Following the recipe in (d), ~~but replaces the DexTrAH-RGB-trained teacher with one trained using~~ we replace the teacher-training pipeline with our customized RL recipe~~(as in (b))~~.

(f) Our RL + VLM-DexSafeDagger. This setting extends (e) by replacing the stationary intervention policy with the proposed VLM tuner, which periodically calibrates the disagreement-risk intervention policy using rollout statistics and representative visual states.

(g) Our RL + LLM-DexSafeDagger. This setting matches (f), except that the language model receives only the rollout statistics and no visual observations.

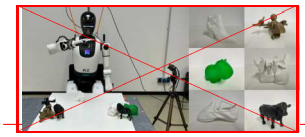
~~Table III shows that ablations~~ **(h) Our RL +** **DexSafeDagger+ Adaptive Intervention under a Performance Curriculum.** We replace the VLM in (f) with an adaptive threshold curriculum inspired by ThriftyDAgger [18] and driven by recent lift-success and unsafe-episode rates, dynamically balancing conservative teacher intervention against aggressive student exploration throughout training. All other components and the threshold-update schedule are held fixed.

To answer question (i), we compare (a) with (b) and (d) ~~, which use the DexTrAH-RGB RL recipe , achieve lower lift success and higher unsafe rates~~with (e). The DexTrAH-RGB-style RL recipe achieves lower performance than our RL recipe. We attribute this gap to the ~~recipe's reliance on a~~ specialized tuning of the FABRICS controller ~~tuned for the KUKA–Allegro platform~~for the original task setup, which transfers poorly to ~~our~~ new hardware and dynamics. ~~The resulting weaker teachers constrain downstream distillation performance.~~

~~As shown~~ We next address question (ii) by examining the distillation behavior in Fig. 3~~, ablations~~. Ablations (b), (c), (e), and (f) achieve comparable final lift success (left), consistent with the intervention policy's focus on safety rather than task success. Unlike Vanilla DAgger ($\beta = 0$), the three intervention variants reduce teacher intervention as the student improves (middle). They reduce early unsafe episodes by approximately 80% relative to Vanilla DAgger and remain below 20% throughout distillation, with VLM-DexSafeDagger achieving the lowest rate overall (right).

~~Comparing ablations (e) and (f) isolates the contribution of VLM-guided adaptation in . Both use the same RL teacher and two-channel intervention policy, but (e) uses fixed thresholds, whereas (f) adapts them using a VLM conditioned on rollout statistics and visual context. Across all evaluated objects, ablation (f) achieves an unsafe-episode rate that is lower than that of (e) by an average of 4 percentage points, while maintaining comparable lift success. This result shows that VLM-guided intervention policy adaptation improves safety without sacrificing task performance.~~

~~Compared with~~ We address question (iii) by comparing ablation (c), which uses disagreement alone, with ablation (e), which adds success-value prediction while retaining the same RL teacher and stationary intervention policy. The combined gate intervenes in low-predicted-success states missed by disagreement alone, reducing the average unsafe-episode rate by about 7.5 percentage points and limiting collision and object-escape failures without sacrificing task performance. Its unsafe rate approaches the teacher's average ($\approx 12\%$) by the end of training and reaches a minimum of approximately 8%.



~~**Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.~~

Fig. 4 further shows that harmful collisions and object escape dominate unsafe episodes. Compared with Vanilla DAgger, all three intervention variants suppress both failure modes, particularly their early spikes, demonstrating that teacher intervention limits unsafe exploration during the initial distillation phase.

Finally, we address question (iv) by comparing ablations (e), (h), (g), and (f), which form a progression from stationary intervention, to performance-curriculum adaptation, to semantics-guided LLM adaptation, and finally to our VLM-guided recipe. This controlled ladder isolates the contributions of feedback-based threshold adaptation, semantic reasoning, and visual context. As shown in Table III, VLM-guided adaptive intervention consistently achieves the highest lift success and the lowest unsafe rate across the evaluated objects. Both LLM-guided and curriculum-based adaptation improve overall safety, but neither matches the full VLM-guided recipe. The curriculum-based intervention also incurs a small reduction in lift success.

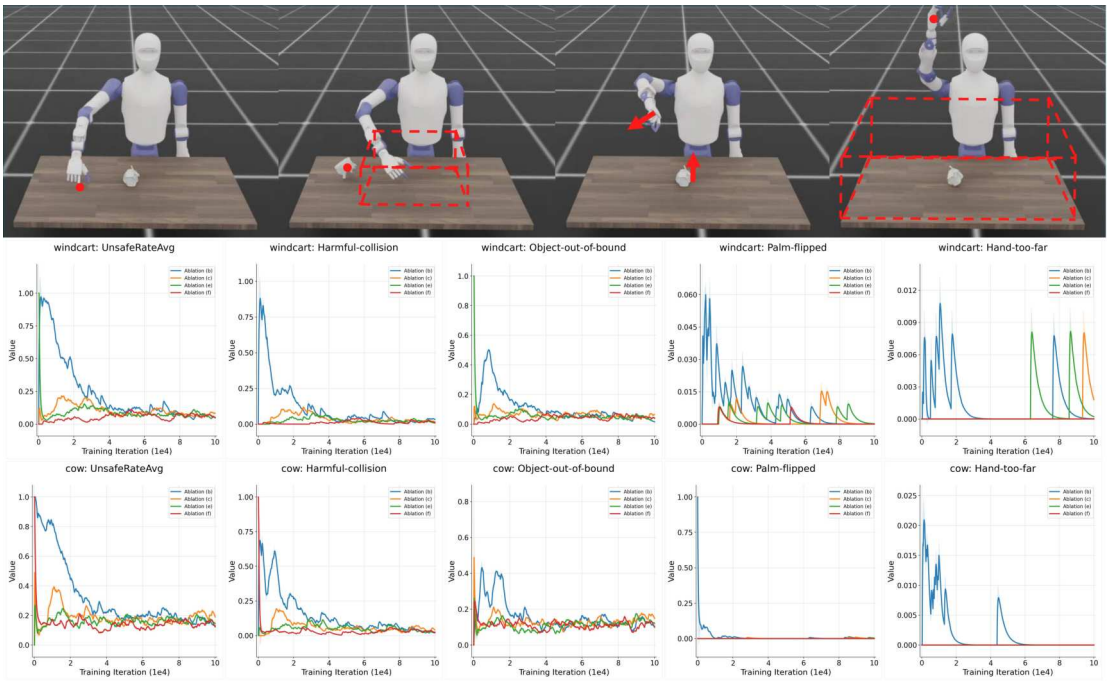


Fig. 4: **Per-object failure-mode decomposition during online distillation ablations.** We illustrate four task-specific failure modes in simulation (harmful collision, object escape, palm flipped, and hand too far) and report the aggregate unsafe rate together with its per-mode breakdown over training iterations. Results are shown for two objects (`windcart` and `cow`); lower is better in all plots. Curves correspond to four distillation variants: blue, Vanilla DAgger [10]; orange, Vanilla SafeDAgger [12]; green, DexSafeDagger; and red, VLM-DexSafeDagger.



Fig. 5: **Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.

D. Real-World Experimental Results

We evaluate the deployed student policy on 22 objects, comprising ~~16~~ 14 seen objects and ~~6~~ 8 unseen objects. Table IV reports a representative six-object subset ~~. The experiments exhibit a lower lift success rate and a higher unsafe episode rate than the teacher and student evaluation results in simulation~~ alongside pooled results over the complete sets of 14 seen and 8 unseen objects. The real-world results show lower lift-success rates and higher unsafe-episode rates than the simulation results, indicating that VLM-DexSafeDagger does not fully eliminate the sim-to-real gap. Nevertheless, the ~~resulting policy is sufficiently robust for real-world deployment~~physical experiments demonstrate that the policy can transfer successfully from simulation to the real robot.

~~The unseen-object evaluation reveals~~ Unseen objects show a clear generalization gap: ~~compared to seen objects (WindCart/Boot/ToolBox/Cow), the policy achieves lower lift success and higher unsafe rates on unseen objects (Plane/Shoe). This degradation is primarily driven by increased~~ pooled lift success drops from 83.6% to 64.6%, while the unsafe rate rises from 14.7% to 39.2%, mainly due to more harmful collisions (~~20.0% and 26.7%on Plane and Shoe) , while object escape remains comparable across objects (≈6.7–20.0~~4.7% to 18.8%) and object escapes (8.1% to 15.8%). ~~No~~ Among the six representative objects shown, no palm-flip or hand-too-far events are observed, suggesting that their real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes.

The real-world comparison between ablations (e) and (f) further shows that our VLM-guided method lowers the unsafe-event rate on both seen (16.1% to 14.8%) and unseen objects (41.4% to 39.2%) while maintaining a comparable task success rate.

V. Conclusion and Future Work

In this work, we propose VLM-DexSafeDagger, which integrates teacher–student disagreement with a Bellman

TABLE IV: **(a):** Representative six-object subset of the 22-object real-world evaluation (~~15~~ 30 episodes per object across seeds 40–44)~~.~~, with pooled results over the complete sets of 14 seen and 8 unseen objects in the last two columns, and **unseen objects** are shaded; **(b):** real-world performance comparison of ablations (e) and (f).

(a)

Metric (%)	WindCart	Boot	ToolBox	Cow	**Plane**	**Shoe**	**Seen avg. (14)**	**Unseen avg. (8)**
Lift Success	90.0	83.3	80.0	86.7	**66.7**	**60.0**	83.6	**64.6**
Unsafe Rate Avg	13.3	16.7	20.0	23.3	**40.0**	**40.0**	14.7	**39.2**
Harmful collision	0.0	3.3	13.3	13.3	**20.0**	**26.7**	4.7	**18.8**
Object escape	13.3	13.3	6.7	10.0	**20.0**	**13.3**	8.1	**15.8**
Palm flipped	0.0	0.0	0.0	0.0	**0.0**	**0.0**	1.4	**3.3**
Hand too far	0.0	0.0	0.0	0.0	**0.0**	**0.0**	0.5	**1.3**

(b)

Metric (%)	Ablation (e)		Ablation (f)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

success-value critic trained on safe-lift-success labels and uses a VLM to adapt the intervention policy. VLM-DexSafeDagger improves exploration safety during multi-teacher distillation, reducing the unsafe-episode rate by **80%** relative to the vanilla DAgger baseline during early training. The introduction of the Bellman critic reduces the average unsafe-episode rate by **7.5 percentage points** relative to the disagreement-only SafeDAgger baseline. In addition, VLM-guided intervention policy adaptation yields consistent improvements across different visual contexts and unsafe scenarios, reducing the average unsafe-episode rate by a further **4 percentage points** relative to DexSafeDagger. The proposed method approaches and, in some cases, surpasses the teacher policy's safety performance. Successful real-world deployment shows that the resulting student policy partially bridges the sim-to-real gap.

A clear limitation is that performance during distillation, as well as the final student policy, is largely constrained by teacher capability. Developing a safety-aware teacher training pipeline is a promising direction for future work to generate stronger teacher policies.

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